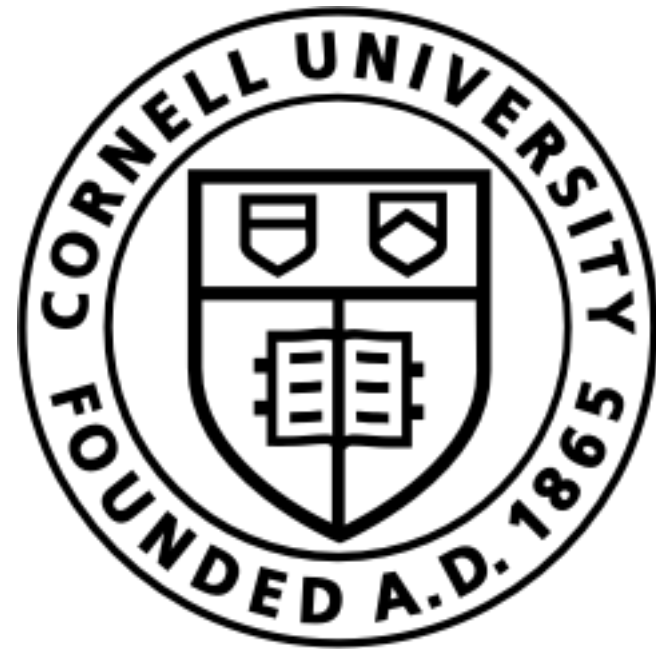




Lecture 5b: Risk and Risk Aversion and Finding My Location on the Capital Allocation Line (CAL)

CHEME 5760 Cornell University

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Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low-risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives. In this lecture, we will:

- ▶ **O1:** Continue with the Insurance Problem, the Certainty Equivalent (CE), and the Risk Premium (RP). How much would you be willing to pay for an insurance policy?
- ▶ **O2:** Revisit the risky + risk-free investment problem. Given your risk attitude, compute the optimal position on the capital allocation line (CAL).

————— next time —————

Introduction to contingencies: Contingencies are a framework to rationalize an uncertain world. We model the randomness of the world and compute using that random model

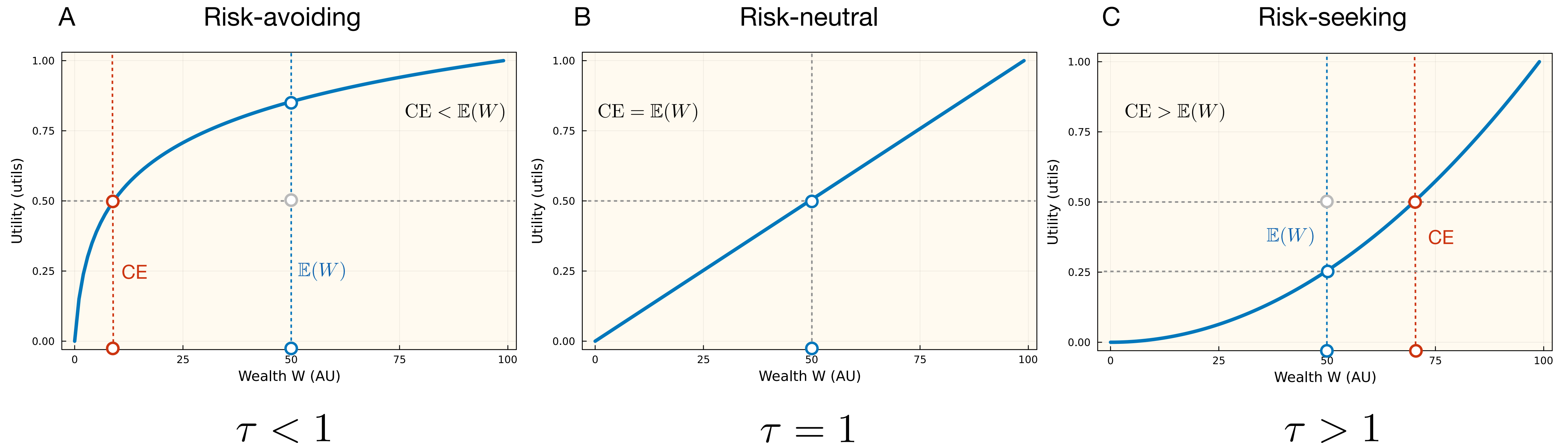
Jupyter Notebooks

Repository: <https://github.com/varnerlab/CHEME-5760-Examples-F23.git>

- Example: CHEME-5760-L5a-InsuranceProblem.ipynb
- Example: CHEME-5760-L5b-CAL-Optimal.ipynb

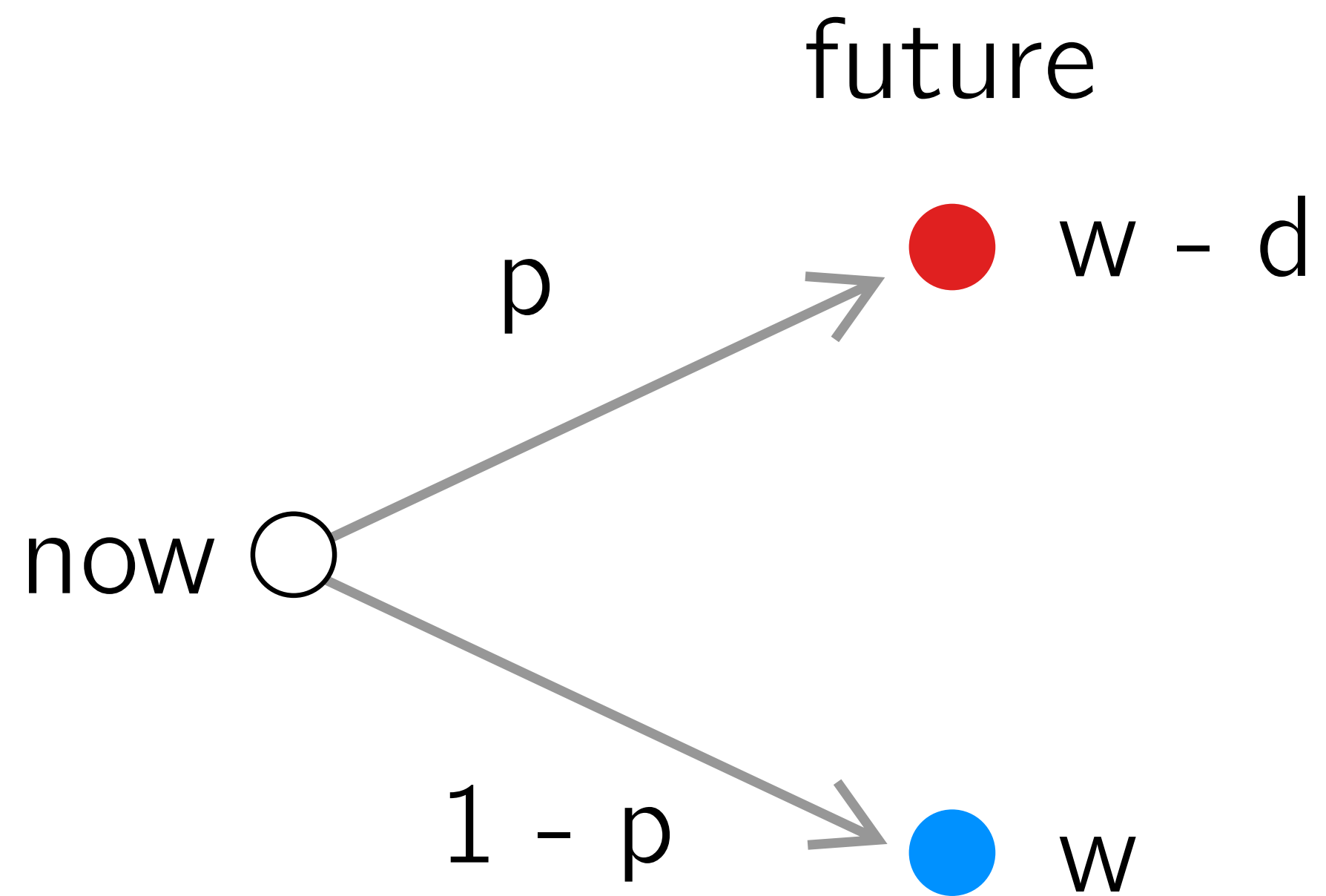
The **certainty equivalent (CE)** is a guaranteed level of wealth that provides the same utility as the expected utility of a risky scenario.

Consider: $U(w, \tau) \sim w^\tau$

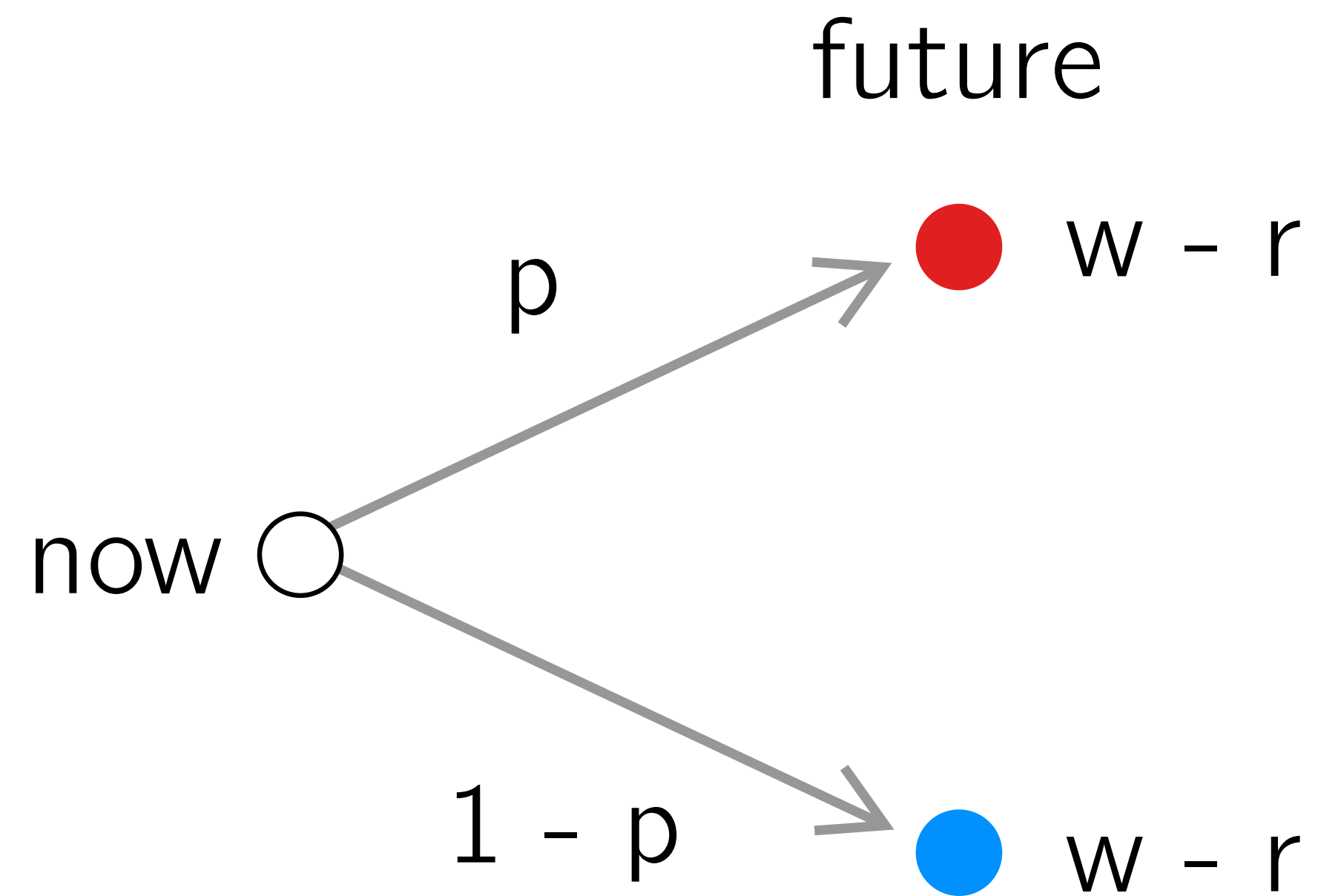


A decision-maker is indifferent between receiving the certainty equivalent and the risky scenario.

Concept: The Insurance Problem



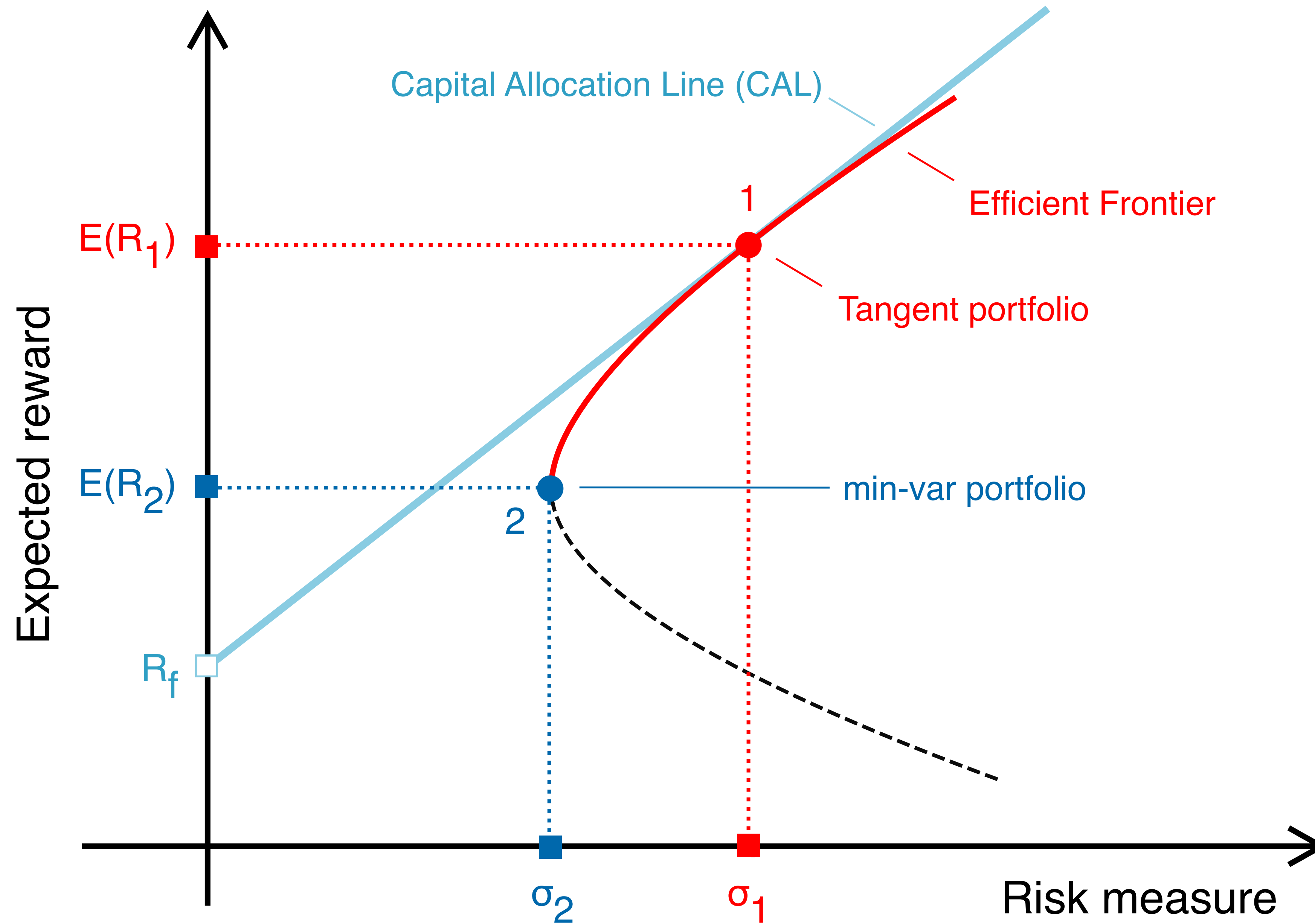
A: without insurance



B: with insurance

Example: CHEME-5760-L5a-InsuranceProblem.ipynb

O2: Revisit the risky + risk-free investment problem. Given your risk attitude, compute the optimal position on the capital allocation line (CAL).



An investor has the following utility function for the complete portfolio $\mathcal{C}$ (risky and risk-free assets):

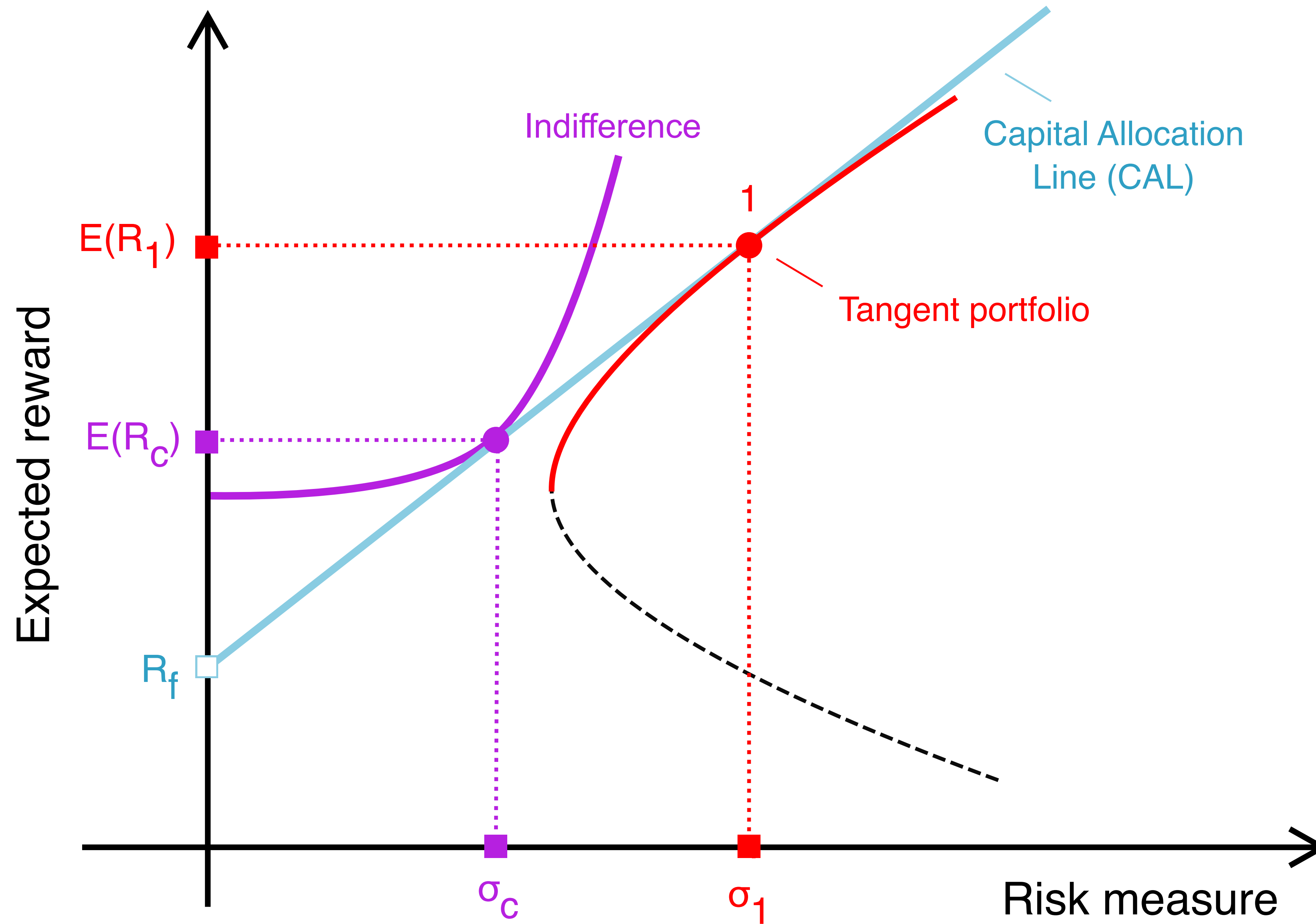
$$U(\mathbb{E}(r_{\mathcal{C}}), \sigma_{\mathcal{C}}) = \mathbb{E}(r_{\mathcal{C}}) - \frac{A}{2} \sigma_{\mathcal{C}}^2$$

where A is a constant that represents the investor's risk aversion. We can rewrite this problem in terms of the risk-free asset and the tangent portfolio $\mathcal{T}$:

$$U(\theta) = r_f + \theta \cdot (\mathbb{E}(r_{\mathcal{T}}) - r_f) - \frac{A}{2} \cdot \theta^2 \cdot \sigma_{\mathcal{T}}^2$$

where θ is the fraction of the investor's wealth invested in the tangent portfolio $\mathcal{T}$, and r_f is the risk-free rate. The optimal θ is:

$$\theta^* = \frac{\mathbb{E}(r_{\mathcal{T}}) - r_f}{A \cdot \sigma_{\mathcal{T}}^2}$$



Example: CHEME-5760-L5b-CAL-Optimal.ipynb

Today?